

1 PHONE STATE AND REQUEST



battery 47%
not charging



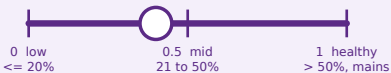
DDR 58 C
battery 35 C



prompt 9.7K tokens
answer length open

2 muKV SCHEDULER (0.3 s per request)

lever L = 0.4 (tier 0.5, loop bias -0.1)



L sets the exchange rate, quality floor, time slack and output cap



ladder walk in time order: take a step only if the energy saved is at least L's exchange rate times the time given up; stop at the time budget or quality floor

plan GPU, prefill 1200 MHz, decode 726

K = 1024 kept answer cap 1024 tokens

3 SNAPDRAGON 8 ELITE GEN 5

Adreno 840 GPU

compute-bound bandwidth-bound

prefill 1200 MHz **decode 726**

cap written as root: max_pwrlevel + max_gpuclk

Oryon CPU

clock: not an energy lever (flat 883 to 1632)

watchdog ladder 1497 / 1382 / 1267 MHz

one step before the vendor's 883

LPDDR: the KV cache

after prefill: 9737 cells, every one scored by the kq_evict side node



compact in place, FA kernel stays on



1024 kept: 4 sink, top-K by score, recent

22.6 MiB live on the GPU; decode reads only this

5 METER AND THE TWO LOOPS (after every request)

USB rail + coulomb counter,
clocks, temperatures;
start-anchored split into
prefill and decode energy

meter



predictions

predicts E and T per plan,
learns from the meter

performance loop

energy loop



time over budget
→ lever +0.1

energy over prediction
→ lever -0.1

both budgets met: the bias decays toward the tier default

bias per tier

4 HEAT HAS ITS OWN PATH



watchdog (CPU): steps the clock down at the battery and skin ladders,
before the vendor limiter

vendor limiter (last resort): GPU to 726 MHz at DDR ~64 C on the cable,
within a minute on battery; our caps only ever sit below it

— control, per request

- - - measurement and learning

- - - heat

■ ■ ■ ■ KV cells: sink, kept, recent, evicted

Measured: 1336 J with the decode cap against 1428 J uncapped on the cable, +1.5% time;
on the real battery 885, 748 and 506 J per request across the three tiers.